

Chun-Che Wen

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Education

Medical University of South Carolina, Charleston, SC August 2019 – August 2024

- Ph.D. in Biostatistics
 - Thesis: Bayesian Beta-Binomial Mixed Models for Longitudinal, Clustered, and Spatiotemporal Data
 - Advisor: Brian Neelon

Florida State University, Tallahassee, FL August 2017 – May 2019

- Master of Science in Statistics

National Chung Hsing University, Taichung, Taiwan September 2012 – June 2016

- Bachelor of Science in Applied Mathematics

Professional Experience

The Dartmouth Institute for Health Policy & Clinical Practice, Lebanon, NH October 2024 – present

- Post-doctoral Fellowship
- Supervisor: James O'Malley

Research Interests

Bayesian inference; changepoints analysis; growth mixture model; multiple comparison; spatiotemporal model; stochastic block model; variable selection; zero-inflated model

Publications

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1. **Wen, C.**, Paul, R., Hunt K., O'Malley A. J., Li, H., Hill, E., Malek, A., Neelon, B. (2026) "A Bayesian zero-inflated spatiotemporally varying coefficients model for overdispersed binomial data," *Journal of Royal Statistical Society, Series A: Statistics in Society, Volume 189, Issue 3*.
 2. Hunt, K., **Wen, C.**, Chundru K., Mateus, J., Florez H., Korte, J., Neelon, B., Wilson, D., Pearce, J., Alkis, M., Simpson, S., Alsbrook, E., Spate, V., Malek, A. (2025), "Adverse Pregnancy and Infant Outcomes by COVID-19 Infection Status before and during pregnancy." *Pregnancy, 1: e70051*.
 3. Malek, A., **Wen, C.**, Neelon, B., Wilson, D., Mateus, J., Pearce, J., Simpson, S., Chundru, K., Korte, J., Florez, H., Hunt, K. (2024) "Prevalence of Hypertensive Disorders of Pregnancy and Chronic Hypertension Increased Throughout the COVID-19 Pandemic in South Carolina (2015-2021)." *Women & Health Journal*, 1–15.
 4. Sun, N., Young, T.C., Wen, M., **Wen, C.**, & Chin, C. (2024) "Stratified Hypothesis Testing of Mean Duration of Response in Randomized Clinical Trials." *2024 Joint Statistical Meetings (JSM), Portland, Oregon. Zenodo*.
 5. Hunt, K., **Wen, C.**, Neelon, B., Wilson, D., Mateus, J., Pearce, J., Simpson, S., Chundru, K., Korte, J., Florez, H., Malek, A. (2024) "Increasing prevalence of diagnosed gestational diabetes in South Carolina: 2015-2021." *Journal of Women's Health (Larchmt)*. 33(11):1518-1527.
 6. Simpson, S, Malek, A., **Wen, C.**, Neelon, B., Wilson, D., Mateus, J., Pearce, J., Chundru, K., Korte, J., Florez, H., Hunt, K. (2024) "Trends in Gestational Weight Gain and Pre-Pregnancy Obesity in South Carolina, 2015 through 2021." *Preventing Chronic Disease 2024;21:240137*.
 7. **Wen, C.**, Baker, N., Paul, R., Hill, E., Hunt, K., Li H, Gray, K., Neelon, B. (2024) "A Bayesian beta-binomial piecewise growth mixture model for timeline followback data." *Statistical Methods in Medical Research*. 33(10):1859-1876.

8. Hung, H-Y., **Wen, C.**, Su, P-F., Man, S., Chang, Y-J. (2024) "Implementation of shared decision-making about rooming-in: A before and after an audit of breastfeeding in Taiwan." *International Breastfeeding Journal*. 19(1), 40.
9. Chundru, K., Korte, J., **Wen, C.**, Neelon, B., Wilson, D., Mateus, J., Pearce, J., Simpson, S., Hunt, K. (2024) "Increasing Preterm Delivery and Small for Gestational Age Trends in South Carolina during the COVID-19 Pandemic" *International Journal of Environmental Research and Public Health*, 21(4):465.
10. **Wen, C.**, Baker, N., Paul, R., Hill, E., Hunt, K., Li H, Gray, K., Neelon, B., (2023). "A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints." *Statistics in Medicine*, 43(1): 125-140.
11. Neelon, B, **Wen, C.**, Benjamin-Neelon, SE (2022). "A multivariate spatiotemporal model for tracking COVID-19 incidence and death rates in socially vulnerable populations." *Journal of Applied Statistics*, 50(8), 1812–1835.
12. Killeen, T., **Wen, C.**, Neelon, B., Baker, N., (2023). "Predictors of Treatment Completion among Women Receiving Integrated Treatment for Comorbid Posttraumatic Stress and Substance Use Disorders." *Substance Use & Misuse*, 58(4), 500–511.
13. Wang, W., **Wen, C.**, Hung, T., Zhong, J., Wen, M. (2022) "Single-stage sampling procedure for heteroscedasticity analysis of means." *Brazilian Journal of Probability and Statistics, Brazilian Journal of Probability and Statistics*, 36(4), 704-724.
14. McGhee, H., Gehle, D., Shope, C., **Wen, C.**, Marston, A., Discolo, C., Pecha, P. (2022) "Feeding Performance and Outcomes in Infants with Robin Sequence Undergoing Mandibular Distraction Osteogenesis." *The Cleft palate-craniofacial journal*, 61(2), 295–301.
15. Wang, W., **Wen, C.**, Das R., Wen, M, (2022). "A modified single-stage sampling procedure for heteroscedasticity analysis of means with unbalanced design." *Communications in Statistics - Simulation and Computation*, 53(5), 2277–2290.
16. Wen, M., **Wen, C.**, Wang, W (2022). "Single-stage sampling procedure for heteroscedasticity in multiple comparisons with a control." *Communications in Statistics - Simulation and Computation*, 53(4), 1719–1728.
17. Yan, F., Levy, D. A., **Wen, C.**, Melvin, C., Ford, M. E., Nietert, P. J., Peach, P. P (2020). "Rural Barriers to Surgical Care for Children with Sleep-Disordered Breathing." *Otolaryngology-head and neck surgery*, 166(6), 1127–1133.
18. Zhou, A., Wang, D., McKee, S., Ye, F., **Wen, C.**, Wallace D. E., Ancell, K. K, Conry, R. M., Johnson, D.B (2019) "Correlates of response and outcomes with talimogene laherperpvec." *Journal of surgical oncology*, 120(3), 558–564.

Under Review

1. **Wen, C.**, Leggett, C., Barnato, A., O'Malley, A. J., A Bayesian spatial hierarchical model with multivariate random-effects for quantifying multi-group regional heterogeneity (under review by *Biostatistics*)
2. Tang, L., **Wen, C.**, Wen, M., Estimating the impact of hedge fund flows on liquidation risk (under review by *International Review of Financial Analysis*).
3. Tomko, R., Dahne, J., Donato, S., **Wen, C.**, Hashemi, D., McClure, E., Gray, k., Wolf, B., Toward personalized medicine for smoking cessation: Predicting individuals' response to smoking cessation pharmacotherapy using multivariable and machine learning models (under review by *Psychological Medicine*).
4. Chen, Y., Huang, H., Tsao, C., Lee, A., Hsu, H., Chen, L., **Wen, C.**, Tai, S., Lee, E., Melatonin supports early synaptic preservation or remodeling after cerebral ischemia: In vitro evidence for MT1/MT2-associated BDNF regulation (under first revision)

Working Paper

1. **Wen C.**, and O'Malley A. J., Bayesian community detection in asymmetric patient-flow network

Presentations

Invited Presentations

2025 A Bayesian zero-inflated spatiotemporally varying coefficients model for overdispersed binomial data. 2025 *Joint*

Statistical Meetings.

- 2024 Exploring the beta-binomial model: Analyzing overdispersed binomial data, *Holling Cancer Center Talk (MUSC)*.
- 2024 A Bayesian beta-binomial piecewise growth mixture model for timeline followback data. *2024 Joint Statistical Meetings*.
- 2023 A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints. *The Eastern Chapter of International Society for Bayesian Analysis (EAC-ISBA) 2023 Conference*.

Contributed Presentations

- 2025 A Bayesian zero-inflated spatiotemporally varying coefficients model for overdispersed binomial data. *ENAR 2025 Spring Meeting*.
- 2024 A Bayesian beta-binomial piecewise growth mixture model for timeline followback data. *2024 WNAR /IMS/Graybill Annual Meeting*.
- 2024 A Bayesian beta-binomial piecewise growth mixture model for timeline followback data. *2024 SC-ASA Palmetto Symposium (Outstanding Presentation Award)*.
- 2024 A Bayesian beta-binomial piecewise growth mixture model for timeline followback data. *ENAR 2024 Spring Meeting*.
- 2023 A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints. *2023 The Association of Clinical and Translational Statisticians (ACTStat) Annual Meeting*.
- 2023 A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints. *ENAR 2023 Spring Meeting*.
- 2023 A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints. *3rd Annual Data Science Workshop (Best 5-Minute Presentation Award)*.

Poster Presentations

- 2023 A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints. *The Southern Regional Council on Statistics (SRCOS)*.

Research Experience

Post-doctoral Fellowship, Geisel School of Medicine at Dartmouth College October 2024-present

PI: James O'Malley

Project: Geographic variations in health care: overcoming bias, expanding generalizability, and improving relevance

Research Assistant, Medical University of South Carolina University August 2021 – August 2024

PI: Kelly Hunt and Angela Malek

Project: Impact of the COVID-19 pandemic, SARS-CoV-2 infection, and social determinants of health on pregnancy complications, birth outcomes and post-pregnancy maternal cardiovascular and mortality outcomes

Research Assistant, Medical University of South Carolina University August 2019 – August 2024

PI: Brian Neelon

Project: Development of multivariate spatiotemporal models for correlated count outcomes and to use these models to track daily COVID-19 rates across all US counties over the course of the pandemic

Summer Research Intern, Vanderbilt University Medical Center May 2018 – July 2018

Supervisor: Fei Ye

Teaching Experience

Dartmouth College

Independent Study Journal Club: Bayesian and Algorithm in Public Health (Spring 2025)

Medical University of South Carolina

BMTRY 719: Bayesian Biostatistics (Fall 2021, Spring 2024)

- Guest lecture for Biostatistics Ph.D. students: Demonstrated an intro and case studies on Bayesian computation using Nimble software.

BMTRY 701: Biostatistical Methods II: Regression Methods in Biology & Medicine (Spring 2023)

- Lab instructor/grader for MPH students: Presented weekly lectures in linear/logistic regressions lectures using SAS software.

BMTRY 700: Biostatistics Methods I: Introduction to Clinical Biostatistics (Fall 2021)

- Lab grader for MPH students: Taught SPSS software and grad homework.

Honors and Awards

- Postdoctoral Scholar Professional Development Award, April, 20205
- Outstanding Presentation Award, SC-ASA Palmetto Symposium, April 2024
- (NSF) Travel Award Winner, 3rd Annual Data Science Workshop, March 2023
- Best 5-Minute Presentation Award, 3rd Annual Data Science Workshop, March 2023
- Travel Award Winner, The Southern Regional Council on Statistics, June 2023

Services

2025	Session Chair at the 2025 JSM
2024	Session Chair at the 2024 ENAR Spring Meetings
2023	SURE (Statistics Undergraduate Research Experience) Program Mentor at 2023 SRCOS
2023	Session Chair at the EAC-ISBA 2023 Conference

Journal Referee

Biometrical; Biostatistics; BMC Medical Research Methodology; Journal of Biopharmaceutical Statistics; Journal of Information Systems Education; Journal of Nonparametric Statistics; Journal of Statistical Theory and Practice; NPJ – Science of Learning; PLOS One; Statistics in Medicine; Statistical Methods in Medical Research (5);

Technical Skills

Statistical Software: R, SAS, Python

Bayesian Software: Nimble, RStan

Other: Rmarkdown, Bookdown, R Shiny